A Multi-Scale Contrast Preserving Encoder-Decoder Architecture for Local Change Detection From Thermal Video Scenes

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***Abstract*— This article presents a new deep-learning architecture based on an encoder-decoder framework that retains contrast while performing background subtraction (BS) on thermal videos. The proposed scheme consists of three consecutive blocks: the encoder, the Multi-Scale Contrast Preservation (MSCP) block, and the decoder. The encoder network employs a hybrid of convolution and atrous convolution blocks to preserve both sparse and dense features, with a skip connection. The encoder, combined with the MSCP block, maintains multi-scale contrast features with reduced training loss. Furthermore, the decoder network accurately projects the extracted features at different layers into pixel-level detail. The proposed end-to-end model efficiently provides a binary map for the corresponding thermal video scene. The efficiency of the proposed algorithm is validated on two large-scale datasets, namely *CDnet 2014* and the *Tripura University Video Dataset at Night Time (TU-VDN)*. Both qualitative and quantitative results demonstrate that MSCP outperforms thirty-eight existing BS schemes.**

***Index Terms*— Deep neural network, background subtraction, multi-scale contrast preservation block.**

I. Introduction

THERMAL sensors can detect long-wave infrared radiation emitted or reflected by objects that are difficult to analyze with normal human vision [1]. Conventional ferroelectric barium strontium titanate (BST) thermal sensors, commonly used in many surveillance cameras, offer a good signal-to-noise ratio (SNR). The development of thermal cameras for surveillance is crucial for various military and naval research endeavors. Thermal-based surveillance systems are primarily used for two major military tasks: Automatic Target Recognition and long-range vehicle detection. However, in recent years, the easy and affordable availability of thermal sensors, along with advancements in surveillance technologies, has led to several other applications in different areas, including industrial safety, fault detection, and night surveillance.

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In computer vision, automatic surveillance using thermal camera videos relies on background subtraction (BS) as one of the essential tools for detecting local changes. For BS, a sequence of image frames is initially used to model the background of a particular video scene. In video sequences, detecting moving objects involves comparing the target image with the established background model. Most BS techniques developed for thermal videos are significantly affected by the low resolution and noise of the video. Furthermore, because the ambient temperature and the body temperature of objects in the video scene are often similar, the scene can exhibit significant camouflaging effects. Other major challenges faced by BS techniques in thermal video include dynamic background conditions, noise level uncertainty, and the multi-level brightness of nearby pixels [2], [3]. These specific challenges require BS methods that are different from those used in conventional surveillance with CCD cameras, which are employed for ground surveillance [4], [5], maritime surveillance [6], [7], [8], [9], and aerial surveillance [10].

The literature review shows that state-of-the-art (SOTA) techniques reported for background subtraction (BS) in thermal video surveillance [4] are rare. In many instances, visual surveillance-based BS techniques are applied to thermal videos. However, since thermal videos are often affected by low resolution or missing detailed information, the SOTA techniques are found to be unsuitable for this purpose. In the literature, multi-scale approaches have been successfully applied to light field image depth estimation [11], anti-illumination image super-resolution [12], and real-time stereo matching [13]. In this context, we have developed a novel encoder-decoder-based multi-scale contrast-preserving deep learning architecture for BS. The proposed encoder network, which incorporates a skip connection, is the first attempt to retain spatial information by considering distinct neighborhood pixels in depth at various levels. The proposed encoder, along with the contrast preservation block, maintains multi-scale contrast features with reduced training loss. Furthermore, the proposed decoder network accurately projects the extracted features at different layers into pixel-level detail. The proposed end-to-end model efficiently provides a binary map for low-resolution thermal image sequences. The main contributions can be summarized as follows:

1) We have developed a unique deep encoder architecture comprising four blocks that incorporates a skip connection, ensuring the preservation of the target scene's detailed features.
2) The distinct outcomes of the four blocks within the deep encoder network make the model robust and generalizable for various challenging datasets.
3) The proposed MSCP module preserves the spatial information of challenging scenes by comprehensively analyzing the neighboring pixels at different levels of depth.
4) The effectiveness of the proposed model is assessed using the leave-one-video-out (LOVO) strategy on unseen data, and it has been found to provide adequate accuracy compared to other existing methods.
5) MSCP achieves higher accuracy than the SOTA methods despite being trained with fewer samples and less computation time, making it quasi-practicable for real-time applications by reaching 24 fps.

The efficiency of MSCP is demonstrated by evaluating it on two popular large-scale datasets, namely *CDnet 2014*[1] and *Tripura University Video Dataset at Night Time (TU-VDN)*.[2] MSCP is compared with thirty-eight SOTA local change detection methods, demonstrating its capacity to offer enhanced accuracy compared to these techniques.

The rest of the article is organized as follows: Section II reviews the literature on object detection in thermal videos. Section III provides an in-depth description of the proposed BS technique. Section IV presents the experiments, including an ablation study. Finally, Section V concludes the article and discusses the scope for future work.

II. State-of-the-Art Techniques

Here, we divide the existing SOTA techniques into three main types: a) Mathematical-based methods, namely statistical parametric techniques [14], [15], statistical non-parametric techniques [16], fuzzy-based techniques [17], [18], and Dempster-Shäfer approaches [19]; b) machine learning-based methods, namely unsupervised subspace learning techniques [20], [21], [22], [23], [24], [25] and supervised deep learning techniques [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37]; and c) signal processing methods, namely Kalman filters [38], [39], sparse models [40], [41], [42], and graph signal processing (GSP) models [43], [44], [45]. In the following sub-sections, we survey the methods primarily developed for thermal videos.

A. Mathematical Based Methods

1) Statistical Parametric Based Techniques: Statistical parametric-based techniques are traditional BS methods that have been used for object detection for the past thirty years [46], [47], [48]. These statistical techniques use parametric models to fit the temporal pixel distributions in a video, and the parameters are estimated using statistical computations [49]. For example, Bhanu and Han [50] proposed an automatic human motion analysis technique for infrared image sequences. Modified least squares allow for the estimation of 3D human walking parameters. Although the motion of the objects is analyzed, this approach is limited by the complexity of parameter estimation.

In 2013, Elguebaly and Bouguila [51] proposed a BS technique for moving objects detection from thermal infrared video scenes where the finite mixtures of multidimensional asymmetric generalized Gaussian distributions are used to model the video data, and Expectation–Maximization (EM) algorithm is used for the parameter estimation. The above analysis concludes that the statistical parametric approaches are overly parameter-dependent and implementation of the same is computationally complex.

2) Statistical Non-Parametric Based Techniques: Many SOTA techniques use non-parametric methods to reduce the computational burden incurred in background modeling [52], [53], [54], [55], [56], [57], [58]. Most commonly, kernel, histogram, and window-based techniques are employed in the SOTA methods for parameter estimation. To emphasize the detection and tracking of small objects in thermal videos, Han et al. [59] proposed a small target detection technique for infrared videos in 2016, where the difference of Gabor (DoGb) filters was proposed and improved to suppress complex background edges for accurate object detection in infrared videos. Furthermore, it was found that ghosts and background clutter impact object detection in infrared videos. In this regard, Qiu et al. [60] developed a BS technique that uses a regional growth strategy to retain moving targets and background areas in thermal videos. In 2024, Kerfa [61] employed an unsupervised Bayesian classifier with a bootstrap Gaussian expectation maximization algorithm. Although non-parametric approaches are less complex, they are affected by lower efficiency for thermal videos.

3) Fuzzy Based Techniques: To handle the challenges of background construction, the majority of techniques explored in the literature employ deterministic approaches. However, it should be observed that changes in succeeding images of the video scenes are relatively regular, and the region experiencing static background modifications (pixel with multi-valued background) is obvious. The multi-valued brightness of pixels in the video results in substantial uncertainty in the spatial and temporal domains. In this context, Maddalena and Petrosino [62] devised a BS scheme where the spatial coherence variant is generated in the fuzzy model to deal with the decision concerns normally appearing from sharp settings and decrease the false alarm problem. Also, it is found that the employment of many-valued background architecture fails to represent the uncertainty within the pixel values. Likewise, using outlier or noisy pixels during the background generation process may result in problems in building a durable background architecture. When the mode of the variation deviates, the presence of a noisy pixel might unduly increase the variability within a specific background category, or it could

[1] http://changedetection.net/

[2] https://mkbhowmik.in/eTuvd.aspx

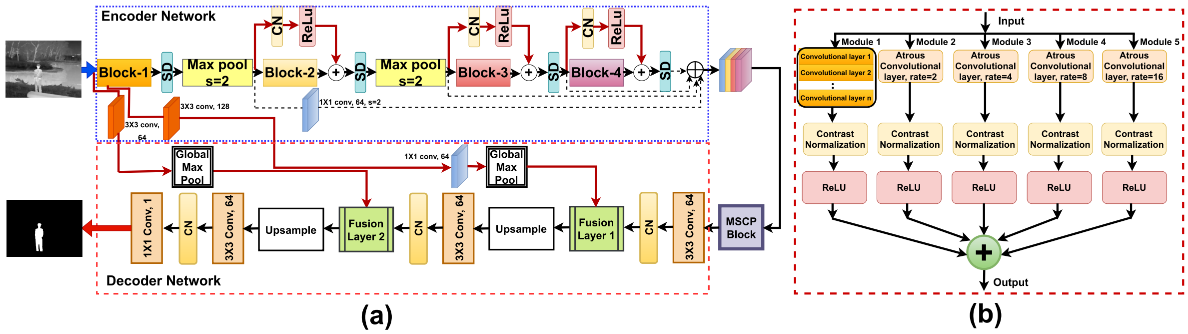

Fig. 1. Block diagram: (a) proposed contrast preserving deep architecture and (b) sample block of the proposed encoder.

attempt to amalgamate multiple background categories into a singular type. In this context, Subudhi et al. [63] developed a background subtraction scheme that leverages fuzzy modal deviation to align pixel values within the background. Introducing kernel-based modal variation facilitates the projection of pixel values into a higher-dimensional space, enabling their distinct split into background and object categories through linear separation.

B. Machine Learning Based Methods

1) Subspace Learning Based Techniques: The use of subspace learning-based techniques for BS is quite popular in the literature. A simple principal component analysis embedded with a fast eigen-decomposition scheme was demonstrated for real-time object detection by Cyganek and M. Woźniak [64]. In 2020, Singha and Bhowmik [65] proposed a BS scheme in which the significant spatial video features are characterized using the Akin-Based Local Whitening Boolean Pattern (ALWBP), which is utilized to separate the foreground and background regions. In another work, Yang et al. [66] developed an integrated low-rank decomposition (CLoD) scheme for local change detection in grayscale-infrared video scenes. This technique detects moving objects as sparse noise compared to the background using a collaborative low-rank structure, and modality weights are employed to achieve adaptive fusion of diverse source data. In 2022, Rai et al. [3] proposed a BS scheme in which relative entropy and Fisher's Linear Discriminant are used for pixel-based categorization and moving object identification in thermal scenes. It should be noted that techniques relying on subspace learning are influenced by significant computational requirements and intricate design complexities.

2) Deep-Learning Based Techniques: With the increasing demand for deep neural network-based techniques, recent SOTA schemes for BS are shifting towards the use of the Convolutional Neural Network (CNN) framework for local change detection [67]. In this context, Yan et al. [68] developed a BS technique in which a deep learning architecture is utilized to train the background model, and a cognition-dependent post-processing algorithm is applied to detect moving objects. It should be noted that most of the existing literature does not incorporate temporal information for object detection. In 2019, Wang et al. [69] developed a BS technique in which the background framework is designed using in-depth and hierarchical spatial-temporal data retained from a multi-scale 3-D fully convolutional network. In addition to the said approaches, several other methods have been reported in the SOTA literature for local change detection in visual surveillance, with applications for thermal videos discussed in three works [70], [71], [72].

Even though these approaches performed well but are not designed specifically for thermal video surveillance, and their accuracy for long-run sequences is yet to be tested. It may be observed that thermal videos are mostly affected by the low resolution or missing information details, the SOTA techniques are found to be not suitable.

Although the leading BS algorithms currently undergoing evaluation on *CDnet 2014* employ deep learning frameworks, their primary limitations concern their computational and memory requirements, and also their supervised aspects requiring labeling of a large amount of data [44], [73], [74]. Furthermore, in the existence of unseen videos, their performance decreases significantly [72], [75], [76]. Thus, we alleviate these three limitations by developing MSCP which requires fewer training samples and less computation time.

III. Proposed Algorithm

We have developed a BS technique consisting of a robust encoder-decoder deep neural network. Detecting moving objects is challenging due to the significant uncertainty in video scenes. As the deep learning network can preserve in-depth features and handle vagueness in challenging scenes, we designed an encoder network with a skip connection that provides better accuracy with increasing depth. The proposed multi-scale contrast preservation block retains discriminative details from the in-depth features and serves as a better feature representation block. The decoder network in the developed model precisely projects the features into image space. A visual illustration of the designed architecture is presented in Fig. 1 (a). The steps of the developed algorithm are outlined as follows: the proposed scheme contains three blocks: the encoder network, the multi-scale contrast preservation block, and the decoder network.

A. The Encoder Network

In this article, we have developed a deep encoder architecture with a skip connection to retain the in-depth features of the target scene. As we know, the depth of a neural network plays an essential role in learning low, mid, and high-level features, and a deep neural network provides improved performance for various computer vision applications. However, if the depth of a CNN increases beyond a certain limit, the accuracy becomes saturated and degrades quickly [77]. To overcome this problem, a skip connection is employed in He et al. [77] to incorporate identity mapping, enhancing the depth of the model while reducing training loss and improving accuracy.

Here, we have developed a novel encoder network with residual connections consisting of four blocks, as shown in Fig. 1 (a). Each block of the encoder framework comprises five modules, from module 1 to module 5, as depicted in Fig. 1 (b). The main motivation behind the proposed encoder is to model the sparse and dense features from the video frames using CNN models. We employed one convolutional layer and four atrous convolutional layers [78] in each block of the designed encoder model. The convolutional layer is responsible for extracting sparse features, while the atrous convolutional layers are used for extracting dense features. The atrous convolution is applied with dilation rates of 2, 4, 8, and 16 to extract more dense features, followed by a contrast normalization (CN) layer [79] and an activation function (ReLU). Module 1 for Block 1 consists of a stack of two 3×3 convolutional layers with 64 filters.

The use of the CN layer in the proposed model achieves an improved performance against batch normalization for smaller batch size [79]. Further, the presence of the ReLU function makes the model faster. In addition to this, module 1 for Block 2 consists of a stack of two 3×3 convolutional layers with 128 filters and a 1×1 convolutional layer. The stacked 3×3 and 1×1 convolutional layers generate a resilient system while utilizing lesser parameters. Further, module 1 for Block 3 consists of a stack of three 3×3 convolutional layers with 256 filters and a 1×1 convolutional layer with 64 filters followed by a CN layer and ReLU. Finally, module 1 of Block 4 consists of a stack of three 3×3 convolutional layers with 512 filters and a 1×1 convolutional layer with 64 filters followed by a CN layer and ReLU.

The proposed encoder architecture is composed of five layers as follows:

1) Convolutional Layer: This layer is utilized to attain the spatial details of the video frame using convolutional kernels. Each convolutional layer has n numbers of kernels of size $K \times K$ to produce various features with size $h \times w \times c$, where h, w, and c denotes the height, width, and depth of the features, respectively. The operation of the convolutional layer can be expressed as

$$FM_j^l = \{I_i^{l-1} * K_j^l + b_j^l\}_{i \in SF_j^l}, \tag{1}$$

where FM_j^l indicates the j^{th} feature map in the l^{th} layer. The term I_i^{l-1} denotes the input image or feature map and K_j^l is the j^{th} kernel in the l^{th} layer. SF_j^l represent the feature map set and b_j^l is the bias. '$*$' indicates the convolutional operation.

2) Atrous Convolutional Layer: At the time of designing the deep model for challenging scenes, we have considered the spatial information of each pixel due to spatial interdependence among the target pixel with its neighborhood pixels. Generally, the spatial details of the scene can be preserved by using a convolution layer with a bunch of kernels. Here, a convolution operation is performed among the kernel and local pixels to generate the sparse feature maps. However, dense feature maps can be obtained from this operation by increasing the receptive field. The receptive field is the size of the region in the input which generates the feature. As a result, the convolutional layer with dilation rate, also known as atrous convolution, is utilized here to extend the receptive field while keeping the training element unchanged. The atrous convolution operation can be represented as

$$K_e = K + (K-1)(\Re - 1), \tag{2}$$

where $K \times K$ denotes the size of filter, the term $\Re$ indicates sampling rate and $K_e \times K_e$ is represents expanded receptive field. In this work, for each block of the proposed encoder, we have employed four branches of convolutional layers with dimensions of 3×3. Within each branch, there are 64 filters incorporated, and these filters possess sampling rates of 2, 4, 8, and 16 respectively.

3) Contrast Normalization Layer: A deep learning architecture can learn the data distribution by training the network. Training the deep neural network is a challenging task since the input distribution of each layer changes at the time of training, as the parameter distribution of the preceding layer change. As a result, the learning rate reduces, leading to a gradual convergence of the model. Consequently, the deep neural network employs the batch normalization layer, leading to an acceleration in training speed enabled by a higher learning rate. Nevertheless, employing contrast normalization in lieu of batch normalization enhances the efficiency of the architecture. The contrast normalization (CN) operation can be represented as

$$CN(O_{ncab}) = \frac{I_{ncab} - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}}, \tag{3}$$

where I_{ncab} indicates the $ncab^{th}$ element of the particular feature map, a and b denotes spatial dimensions, feature map channel is represented by c and the term n is the index of the feature map. The outcome of the CN layer is O_{ncab}.

4) Activation Layer: The linearity of the convolutional operation can constrain the effectiveness of a deep neural network. To address this, an activation function is introduced in a convolutional layer, which imparts non-linearity to the network and enhances its ability to learn the challenging scenes. Thus in the proposed encoder network, we have employed the rectified linear unit (ReLU) as the activation function. This choice not only imparts learned parameters to values exceeding zero but also contributes to the network's speed and efficiency. The ReLU function R can be defined as

$$R(Y) = \begin{cases} Y & Y > 0 \\ 0 & Y \le 0 \end{cases}. \tag{4}$$

5) Spatial Dropout Layer: The introduction of a dropout layer in a deep neural network can help to avoid overfitting and enhance learning outcomes. Since our proposed encoder model is a fully convolutional neural network and the images are strongly spatially correlated, the dropout layer fails in this case. Therefore, in the proposed model, we have used a spatial dropout layer to drop the entire feature map at a rate of 0.25 instead of dropping individual neurons from the feature map. It is observed that when the neighboring pixels have strong spatial inter-dependency in the feature map, the spatial dropout layer provides improved learning performance compared to the standard dropout layer.

In the proposed model, we have utilized all the aforementioned layers to design a robust encoder network that provides better multi-scale features at low, mid, and high levels. The use of both convolutional and atrous convolutional layers in the encoder network helps retain the sparse and dense details of the input image. Similarly, the inclusion of the CN layer in the proposed model achieves improved performance compared to batch normalization for smaller batch sizes [79]. In addition, the presence of ReLU makes the model faster. In the developed model, no form of data pre-processing strategy is utilized for the encoder input. The proposed encoder network contains four blocks, and each block is composed of five modules.

For the same input with the dimension of $H \times W$, the distinct outcomes of all the modules are added and driven from Block1 to Block2 through the spatial dropout (SD) layer [80] with a rate of 0.25 and a max-pooling layer. Again, the same is propagated from Block 1 to Block 2 through a skip connection followed by CN and ReLU layer and combined with the output of Block 2. Similarly, the data transferred is accomplished from Block 2 to Block 3 but Block 3 to Block 4 is done without a max-pooling layer. The skip connection in the proposed model transfers the fine details of the image from the lower block to the higher block, producing coarse details of the image. The amalgamation of fine and coarse details has the potential to elevate the representation of features and ultimately enhance the effectiveness of the proposed framework. To decrease the computational complexity of the model, we applied max-pooling layers after the SD layer in both Block 1 and 2, which effectively reduced the spatial dimension of input features. The max-pooling layer utilizes a filter size of 2×2 and a stride of $s = 2$. Finally, the outputs of Block 1 are passed through a 1×1 convolutional layer comprising 64 filters. Subsequently, the outputs of Block1, Block 2, Block 3, and Block 4 are concatenated along the channels that correspond to the feature maps originating from the encoder (F) with size $\frac{H}{4} \times \frac{W}{4} \times 192$ and can be expressed as follows:

$$F = EB1 \oplus EB2 \oplus EB3 \oplus EB4, \tag{5}$$

where $EB1$, $EB2$, $EB3$, and $EB4$ denotes the output of four different encoder blocks. The symbol $\oplus$ denotes the concatenation utilized to combine the outputs generated by the four blocks of the encoder network.

B. The Multi-Scale Contrast Preservation Block (MSCPB)

The interconnection of pixels within video scenes is crucial for detecting the objects in motion, as a deficiency in this relationship can result in numerous isolated points being incorrectly identified as false alarms within the outcome. Hence, in this developed architecture, we have developed a new multi-scale contrast preservation (MSCP) module as shown in Fig. 2 to retain the relation among the pixels in the challenging video scenes. Also, The MSCP block preserves the multi-scale sparse and dense features accurately from the target frame which are essential for the detection of objects in motion. The deep features extracted from the encoder network (F) with size $\frac{H}{4} \times \frac{W}{4} \times 192$ are given as input to the MSCP module that is present between the encoder network and decoder network. The MSCP block is the hybridization of a max-pooling layer with 1×1 convolutional layer having 64 filters, a 3×3 convolutional layer, and different atrous convolutional layers with sampling rate 4, 8, 16 with each consisting of 64 filters followed by kernel size 3×3 avg-pooling layer. For the encoder's deep feature (F), the max-pooling layer is used to retain the prominent detail in every pooling area and the output of this layer is followed by 1×1 convolutional layer and ReLU function represented by $\mathcal{M}$. As the identification of moving objects is a binary classification task, the findings exhibit a significant contrast between the moving object and the background. Hence, in this work, we have preserved the contrast details $\mathcal{X}_j$ of the encoder's feature and can be given as

$$\mathcal{X}_j = ReLU\{CV_j - AvgPool(CV_j)\}, \tag{6}$$

where CV_j denotes the feature maps of a convolutional and various atrous convolutional layers, $j \in \{1, 2, 3, 4\}$.

The output of the MSCP block ($\mathcal{Y}$) is generated by concatenating $\mathcal{M}$ and $\mathcal{X}_j$ along the depth dimension and processed through the CN layer followed by ReLU and an SD layer with a rate of 0.25. The application of contrast normalization aids in stabilizing the training strategy by diminishing internal covariate shifts, which in turn enhances the model's performance through improved generalization. Additionally, using ReLU succeeded by the SD layer with a 0.25 rate improves the learning efficacy of the developed framework, even with lesser training data. The feature maps at different scales originating from the MSCP block ($\mathcal{Y}$) with size $\frac{H}{4} \times \frac{W}{4} \times 240$ can be computed as follows:

$$\mathcal{Y} = SD\{CN\{\mathcal{M} \oplus \mathcal{X}_j\}\}. \tag{7}$$

In the BS method, accurately identifying objects under challenging conditions is essential for robust performance.

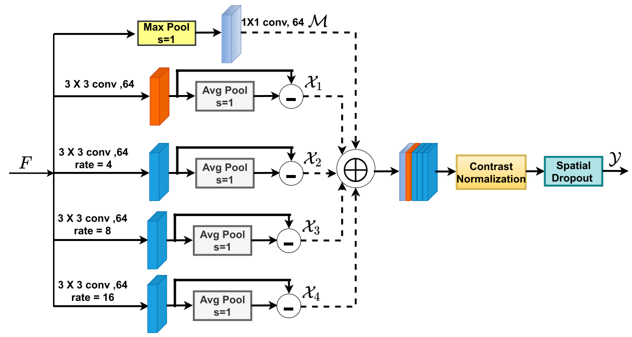

Fig. 2. Graphical demonstration of the developed multi-scale contrast preservation block (MSCPB).

As a result, the multi-scale contrast preservation (MSCP) block is a crucial component in BS algorithms because it helps preserve the contrast of objects across multiple scales. Objects in a scene can vary in scale due to factors like distance from the camera or inherent size differences. The MSCP block addresses these variations by simultaneously analyzing the scene at multiple scales, ensuring that objects of different sizes are effectively detected. Additionally, maintaining contrast between objects and their backgrounds is crucial for accurate object detection. The MSCP block specifically focuses on preserving this contrast across different scales, ensuring that objects remain distinguishable even in challenging lighting conditions or complex backgrounds.

Furthermore, BS algorithms can be susceptible to noise, leading to false detections or missed objects. By preserving the contrast of moving objects, the MSCP block helps mitigate the impact of noise, resulting in more robust and reliable detection outcomes. Additionally, the MSCP block contributes to overall detection accuracy by enhancing the visibility of moving objects while suppressing irrelevant information in the scene. This leads to more precise localization of objects in motion.

Therefore, the MSCP block serves as a key component in the proposed algorithm by addressing scale variation, preserving contrast, reducing noise, and improving overall detection accuracy. This enhancement boosts the performance and reliability of these systems in various real-world scenarios, including night video surveillance, traffic monitoring, and autonomous driving.

C. The Decoder Network

The decoder framework is comprised of 3 blocks, in which each block encapsulates a 3×3 convolutional layer with 64 filters, a CN layer, and a ReLU activation function. The features at multi-scale from the MSCP block with size $\frac{H}{4} \times \frac{W}{4} \times 240$ are sent into the first block of the decoder architecture, which generates with size $\frac{H}{4} \times \frac{W}{4} \times 64$ feature maps. This block's output is fused with low-level features retained from the end of module 1 of the encoder's first block using a 3×3 convolutional layer consisting of 128 filters. These low-level details are transferred from the encoder to the decoder network via a skip connection, which is followed by a 1×1 convolutional layer, 64 filters, and a global Max Pooling layer. The fusion layer 1 in the decoder enhances the feature representation where the low-level features are initially multiplied with the output of the decoder's first block and subsequently combined with the initial feature maps obtained by the same. Finally, these fused feature maps are up-sampled with the dimensions of $\frac{H}{2} \times \frac{W}{2} \times 64$ and given to the next block. Here, a similar kind of operation is performed, and fusion layer 2 boosts the feature representation. Here, for fusion, the low-level features are extracted using a 3×3 convolutional layer with 64 filters from the beginning of the encoder's first block and propagated towards the decoder followed by skip connection and global Max Pooling layer. The fused feature maps from fusion layer 2 are up-sampled with the dimensions of $H \times W \times 64$ and fed to the third block of a decoder, which projects the fused features into 64 feature maps. The output of this block is followed by a 1×1 convolutional layer with 1 filter and a sigmoid function that provides a single feature map $H \times W$. We have applied a threshold value of 0.9 on the single feature to predict, each pixel is either in the objects or in the background.

IV. Experimental Results and Discussions

The developed model is enforced on a Windows 10 system with a Core $i7$ processor and 16 GB of RAM, using Python. Training and testing were conducted on an *NVIDIA* Tesla $P100$ GPU provided by Google Colab. The developed model utilizes the $TensorFlow$ backend with the $Keras$ library.

A. Parameter Settings With Training Details

The designed architecture's training is performed end-to-end using a $NVIDIA$ Tesla $P100$ GPU system in which the batch size $= 1$. Using smaller batch sizes introduces additional noise into the training process, serving as a form of regularization to prevent over-fitting by discouraging heavy reliance on specific patterns within the training data. Moreover, smaller batch sizes often accelerate convergence during training since weight updates occur more frequently, enabling the model to promptly adjust its parameters in response to the training data. Additionally, training with smaller batch sizes consumes less memory compared to larger batch sizes. Furthermore, models trained with smaller batch sizes typically exhibit improved generalization to unseen data. This improvement stems from the noise introduced by smaller batches, which encourages the model to acquire more robust and generalizable data representations.

The training of the developed model on the *CDnet 2014* database containing $N = 200$ frames. The *CDnet 2014* benchmark database is chosen due to its comprehensive coverage of diverse variations and complexities of video scenes found in real-world scenarios. Additionally, it offers high-quality annotations, which are essential for supervised learning tasks. Furthermore, the dataset is openly accessible to researchers, promoting reproducibility and facilitating comparison between different SOTA methods. To assess the effectiveness of the proposed architecture, all frames from diverse categories within the *CDnet 2014* database have been taken into consideration. For training the proposed scheme, we employed the *RMSProp* optimizer with parameters $\epsilon = 1e - 08$ and $\rho = 0.9$. This results in a quicker convergence rate when contrasted with traditional optimizers. *RMSProp* is a stochastic gradient descent technique explicitly designed to enhance the generalization capability of neural networks, mitigating model over-fitting. A learning rate of 0.0001 is initially assigned. Employing a smaller learning rate during the training phase of the developed model can result in enhanced optimization stability, improved generalization, and smoother convergence. The learning rate is decreased by a factor of 10 if the validation loss does not decrease after 5 consecutive epochs. The model is trained for a maximum of 100 epochs, but an early stopping mechanism is employed if the validation loss fails to improve for 10 consecutive epochs. Sequentially feeding training frames into the model could result in biased learning weights due to the strong correlation between successive frames. To address this, training frames are randomly selected before training begins. These frames are then divided into 90% for training and 10% for validation. To tackle the issue of imbalanced data classification, higher weights are assigned to the foreground class, while lower weights are given to the background class. The class imbalance issue arises when certain classes have significantly more instances than others, causing standard classifiers to prioritize the dominant classes and overlook the smaller ones. In the context of foreground segmentation training, background pixels often outnumber foreground pixels. To address this, we calculated class weights adjusting them proportionally to the inverse of class frequencies for both background and foreground classes. This approach assigns higher weights to the rare foreground class and lower weights to the prevalent background class during training, thus mitigating data imbalance. We computed class weights independently by considering the foreground/background pixel ratio in each training frame.

The binary cross-entropy loss function ($BCEL$) is particularly effective when the target variable is binary (0 or 1) and the output of the model is a probability value representing the likelihood of a particular class. Also, $BCEL$ acts as a form of regularization, helping to prevent over-fitting by penalizing large deviations from the true distribution. This property is particularly useful when the data is imbalanced or when there are outliers. Hence, training the model involves the utilization of the binary cross-entropy loss ($BCEL$) function, outlined as follows:

$$BCEL = -\frac{1}{n}\sum_{a=1}^{n}[y_a log(\hat{y}_a) + (1 - y_a)log(1 - \hat{y}_a)]. \quad (8)$$

Here, n represents the total of pixels in the training image, $y_a \in \{0, 1\}$ corresponds to the true label of pixel a, and $\hat{y}_a$ denotes the predicted label for that pixel.

Similarly, for the *Tripura University Video Dataset at Night Time (TU-VDN)* dataset, to train the proposed model, we kept all the parameters the same as mentioned above. Nonetheless, we employed a training set comprising $N = 100$ frames and subsequently evaluated the model's performance using the entire frames. The *TU-VDN* database is chosen due to its extensive coverage of a variety of adverse environmental conditions including low-light, rainy, dusty, and foggy conditions, with challenging problems such as flat cluttered backgrounds and non-static backgrounds found in real-world scenarios. Also, it provides better annotations that are necessary for supervised learning tasks.

B. Analysis of Results

The developed framework is evaluated on two databases: *CDnet 2014* [81], and *Tripura University Video Dataset at Night Time (TU-VDN)* [65]. To evaluate the efficacy of MSCP, we conducted a comprehensive analysis encompassing both visual and quantitative evaluations comparing the results obtained using MSCP to those obtained by thirty-eight SOTA methods.

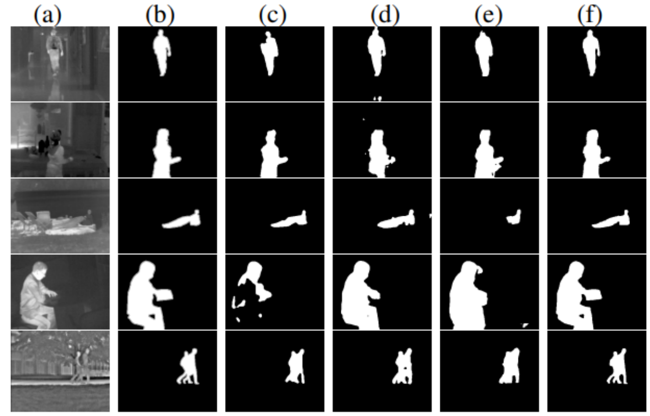

Fig. 3. Local change detection: (a) Input frame (b) Ground-truth frame, Foreground masks (c) DeepBS, (d) Cascade CNN, (e) BSUV-Net and (f) **MSCP (Proposed)**.

1) CDnet 2014 Dataset: It is a well-known benchmark database for validating several moving objection detection techniques. This database includes (a 2014 dataset) with 11 various categories, each with around 4 to 6 video scenes. This database comprises a variety of complex categories such as non-static background, baseline, camera jitter, low frame rate, intermittent object motion, thermal, shadows, adverse weather conditions, PTZ, night video, and air turbulence. The dataset includes 53 video sequences, encompassing nearly $\sim 1, 60, 000$ frames. For each frame, this database comprises manual annotation segmented ground-truth foreground frames.

a) Visual analysis: The visual assessment of moving object detection is performed using all the thermal video scenes from the *CDnet 2014* dataset, as showcased in Fig. 3. The input images and the associated ground-truth frames from the *CDnet 2014* dataset are presented in Fig. 3 (a)-(b). The object detection outcomes attained by the DeepBS scheme are presented in Fig. 3 (c), which has a high false-positive rate. Fig. 3 (d) illustrates the object detection outcomes achieved by the Cascade CNN scheme, in which the pixels in the background are incorrectly referred to as moving object pixels. The findings of the BSUV-Net scheme are depicted in Fig. 3 (e), where numerous false alarms occur in the image, resulting in over-segmentation. However, the outcomes achieved by the proposed scheme are illustrated in Fig. 3 (f), confirming our findings by demonstrating better results compared to SOTA techniques.

b) Quantitative analysis: Thermal sequences available in the *CDnet 2014* dataset suffer from various challenges, including low contrast, high noise, adverse environmental conditions, occlusions, poor texture, class imbalance, and thermal reflection. Hence, for the *CDnet 2014* dataset, we have used four quantitative measures: average precision (PC), average recall (RE), average F-measure (FM), and the average percentage of wrong classifications (PWC) [63]. The precision measure is crucial in moving object detection, where the cost of false positives is high. High precision indicates that the detection method is effective at minimizing false positives. Additionally, recall is critical in situations where missing a detection (false negative) is costly. High recall ensures that

TABLE I
QUANTITATIVE COMPARISON ON FIVE THERMAL SEQUENCES (*CDnet 2014*)

	Algorithms	PC	RE	FM	PWC
Non-deep learning	Fuzzy Mode [63]	0.9344	0.9766	0.9550	1.1130
	KDE [54]	0.8974	0.6725	0.7423	1.6795
	GMM [49]	0.8652	0.5691	0.6621	4.2642
	PAWCS [55]	0.8280	0.8504	0.8324	1.4018
	SuBSENSE [57]	0.8328	0.8161	0.8171	2.0125
	SOBS-CF [62]	0.8715	0.6347	0.7140	1.8021
	WeSamBE [56]	0.8554	0.7727	0.7962	2.3538
	Multimode Background [48]	0.8268	0.8162	0.8194	1.4289
	SharedModel [47]	0.8072	0.8618	0.8319	1.8656
	Spectral-360 [46]	0.9114	0.7238	0.7764	1.6337
Deep learning	DeepBS [70]	0.9257	0.6637	0.7583	3.5773
	WisenetMD [52]	0.8696	0.7867	0.8152	1.8993
	Cascade CNN [71]	0.8577	0.9461	0.8958	1.0478
	IUTIS-5 [53]	0.8969	0.7990	0.8303	1.1484
	BSUV-Net [72]	0.8551	0.8739	0.8581	1.7058
	SemanticBGS [58]	0.9118	0.7664	0.8219	1.3897
	BSUV-Net 2.0 [83]	0.9359	0.8594	0.8932	1.1659
	MSCP (Proposed)	**0.9718**	**0.9858**	**0.9787**	**0.5485**

TABLE II
QUANTITATIVE COMPARISON WITH ALL THE SEQUENCES (*CDnet 2014*)

	Algorithms	PC	RE	FM	PWC
Non-deep learning	Fuzzy Mode [63]	0.8912	0.8672	0.8792	0.4798
	KDE [54]	0.5811	0.7375	0.5688	5.6262
	GMM [49]	0.6025	0.6846	0.5707	3.7667
	PAWCS [55]	0.7857	0.7718	0.7403	1.1992
	SuBSENSE [57]	0.7509	0.8124	0.7408	1.6780
	SOBS-CF [62]	0.5831	0.7805	0.5883	6.0709
	WeSamBE [56]	0.7679	0.7955	0.7446	1.5105
	Multimode Background [48]	0.7382	0.7389	0.7288	1.2614
	SharedModel [47]	0.7503	0.8098	0.7474	1.4996
	Spectral-360 [46]	0.7054	0.7345	0.6732	2.2722
Deep learning	DeepBS [70]	0.8332	0.7545	0.7458	1.9920
	WisenetMD [52]	0.7668	0.8179	0.7535	1.6136
	Cascade CNN [71]	0.8997	**0.9506**	**0.9209**	0.4052
	IUTIS-5 [53]	0.8087	0.7849	0.7717	1.1986
	BSUV-Net [72]	0.8113	0.8203	0.7868	1.1402
	SemanticBGS [58]	0.8305	0.7890	0.7892	1.0722
	BSUV-Net 2.0 [83]	0.9011	0.8136	0.8387	0.7614
	MSCP (Proposed)	**0.9185**	0.9197	0.9185	**0.1177**

most relevant objects are detected, minimizing false negatives. The F-measure is especially useful when there is a need to balance precision and recall. It is an effective measure when the cost of false positives and false negatives is approximately equal or when both aspects are critical. It ensures that neither high precision nor high recall alone is driving the performance evaluation.

Furthermore, the percentage of wrong classifications measures the proportion of incorrect classifications (both false positives and false negatives) out of the total number of classifications. In real-time applications, a high percentage of wrong classification rates can lead to significant practical issues. For example, in an autonomous surveillance system, a high PWC rate might result in frequent false alarms or missed detections. Therefore, minimizing the PWC rate is critical for the operational effectiveness and safety of the detection system. Hence, a higher value of precision, recall, and F-measure, alongside a lower value of the percentage of wrong classifications, indicates the effectiveness of any BS technique [18], [82]. Furthermore, the F-test and P-test are conducted on the *CDnet 2014* dataset for statistical analysis between the proposed and SOTA methods. The outcomes of the proposed technique on thermal scenes (5 videos) are reported in Table I. The developed technique is compared against seventeen existing BS techniques. It is found that the developed scheme provides higher accuracy in terms of all considered measures. We also evaluated the proposed method on all accessible videos (53 videos) and found that it surpasses the seventeen SOTA schemes (as shown in Table II). Table III demonstrates the effectiveness of MSCP using several other evaluation measures. Table IV compares the run-time of the proposed scheme against the most popular deep neural network-based SOTA algorithms tested on the *CDnet 2014* dataset. The processing time required by the proposed model is 24 frames per second, which is faster than the SOTA deep learning-based methods.

2) TU-VDN Dataset: The *TU-VDN* dataset contains various outdoor nighttime video scenes captured by FLIR-t650sc thermal sensors across four different categories: dust, fog, rain, and low light. The image sequences are taken in a variety of adverse environmental conditions, including rainy, dusty, and foggy situations, with challenging issues such as flat cluttered backgrounds and non-static backgrounds under a static sensor.

a) Visual analysis: The visual assessment of local change detection is performed using four testing sequences from the *TU-VDN* database, as illustrated in Fig. 4. The input images and the corresponding ground-truth images from the *TU-VDN* dataset are presented in Fig. 4 (a)-(b). The object detection outcomes obtained by the SuBSENSE scheme are shown in Fig. 4 (c), where many object pixels are misclassified as background. Fig. 4 (d) represents the foreground masks obtained by the LOBSTER method, which generates holes and missed alarms. The foreground masks of the PBAS technique are shown in Fig. 4 (e), where false detections are observed. Nevertheless, the foreground masks of MSCP, shown in Fig. 4 (f), accurately detect all object pixels with enhanced precision.

b) Quantitative analysis: Thermal sequences available in the *TU-VDN* dataset suffer from adverse environmental conditions, including low light, rain, dust, and fog, as well as challenging issues such as flat cluttered backgrounds and non-static backgrounds. In scenarios characterized by low light, rain, dust, and fog, the F-measure excels by striking a balance between precision and recall, effectively capturing the trade-off between correctly identifying objects and minimizing misclassifications in conditions of reduced visibility and increased noise levels. Meanwhile, the Matthews correlation coefficient offers a nuanced assessment by considering all elements of the confusion matrix, making it resilient to imbalanced class distributions and complex background structures. Additionally, accuracy provides a straightforward indication of overall correctness, complementing the F-measure and Matthews correlation coefficient to offer a holistic view of the detection system's performance. Therefore, the developed algorithm is evaluated using three quantitative measures: average F-measure (FM), average Matthews correlation coefficient (MCC) [84], and average accuracy (ACC) [65] for the *TU-VDN* dataset. The results of the devised scheme's performance on this database are elaborated in Table V. We have used fifteen SOTA techniques for evaluation of the devised scheme on *TU-VDN* dataset. It can be observed that except accuracy measure for fog sequence, MSCP produces a higher accuracy by all measures on various categories against the SOTA techniques.

C. Ablation Study

Ablation studies were carried out to assess the efficacy of different components within the proposed technique. For this study, we replaced different considered components with nearby competing technologies to see the efficiency

TABLE III
QUANTITATIVE ANALYSIS OF MSCP (*CDnet 2014*)

Category	Recall	Specificity	FPR	FNR	Precision	F-Measure	PWC
BadWeather	0.9753	0.9996	0.0004	0.0247	0.9668	0.9710	0.0758
Baseline	0.9966	0.9989	0.0011	0.0034	0.9822	0.9892	0.1144
Camera Jitter	0.9927	0.9997	0.0003	0.0073	0.9936	0.9932	0.0596
Dynamic Background	0.9935	0.9999	0.0001	0.0065	0.9896	0.9915	0.0134
Intermittent Object Motion	0.9915	0.9997	0.0003	0.0085	0.9966	0.9940	0.0691
Low Framerate	0.7427	0.9998	0.0002	0.2573	0.7473	0.7448	0.0532
Night Videos	0.9641	0.9994	0.0006	0.0359	0.9681	0.9660	0.1226
PTZ	0.7424	0.9999	0.0001	0.2576	0.7456	0.7440	0.1707
Shadow	0.9954	0.9998	0.0002	0.0046	0.9956	0.9955	0.0365
Thermal	0.9858	0.9941	0.0059	0.0142	0.9718	0.9782	0.5485
Turbulence	0.7371	0.9999	0.0002	0.2629	0.7461	0.7360	0.0311
Average	0.9197	0.9992	0.0009	0.0803	0.9185	0.9185	0.1177

TABLE IV
RUN-TIME (*CDnet 2014*)

Techniques	Run-time frames per second (fps)
DeepBS [70]	10
WisenetMD [52]	12
Cascade CNN [71]	13
BSUV-Net [72]	6
SemanticBGS [58]	7
MSCP (Proposed)	**24**

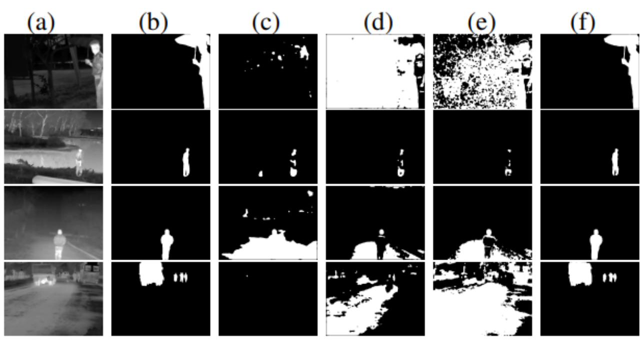

Fig. 4. Local change detection for various video scenes: (a) Input frame (b) Ground-truth frame, foreground masks obtained by (c) SuBSENSE, (d) LOBSTER, (e) PBAS and (f) **MSCP (Proposed)**.

of the devised algorithm. In the initial stage of the experiment (AS1), we performed the BS scheme with the proposed encoder-decoder configuration with a feature pooling module (FPM). The feature pooling module contains a max-pooling layer combined with a 1 × 1 convolutional layer including 64 filters, and a 3 × 3 convolutional layer with 64 filters. Furthermore, this FPM module integrates numerous atrous convolutional layers, each employing 64 filters and having sampling rates of 4, 8, and 16. The features extracted from the encoder output F are subjected to a max-pooling layer and then passed through a 1 × 1 convolutional layer containing 64 filters. These resulting features are denoted as F_1. The features emanating from the encoder's output are subjected to a convolutional layer using a 3 × 3 kernel and 64 filters, denoted as F_2. Then, F and F_2 are concatenated and successively given to the atrous convolutional layer with a sampling rate of 4 to generate the F_3. Similarly, the features F_4 and F_5 are generated from the atrous convolutional layers with a sampling rate of 8 and 16, respectively. Finally, all the features are concatenated, followed by the CN layer, ReLU, and SD layer representing the output of the FPM block. In the following stage of the experiment (AS2), we put forth an encoder-decoder deep-learning architecture that incorporates a multi-scale contrast preservation (MSCP) block, desired to detect local changes. Within the encoder framework, every block consists of three modules: module 1, module 2, and module 3. Each block consists of various convolution and atrous convolutional layers. For this experiment, the atrous convolution layers are considered to be with dilation rates of 2, and 4 followed by a CN layer and a ReLU function, respectively. Afterward, we conducted another experiment (AS3) where each block of the encoder network comprises four modules: module 1, module 2, module 3, and module 4. For this experiment, the atrous convolution layers are considered to have dilation rates of 2, 4, and 8, followed by a CN layer and a ReLU function, respectively. Furthermore, an additional experiment (AS4) is conducted, wherein pre-trained deep neural architectures like VGG-16 and ResNet-50 are employed as encoders, accompanied by a proposed multi-scale contrast preservation (MSCP) block and decoder network to detect local changes for the target scene. At the final stage of the experiment (Proposed), we have developed a novel encoder-decoder deep-learning framework with an MSCP block for moving object detection. Each block of the encoder network comprises five modules: module 1, module 2, module 3, module 4, and module 5. Every block is composed of four atrous convolutional layers along with multiple convolutional layers. The atrous convolution layers are considered to have dilation rates of 2, 4, 8, and 16, followed by a CN layer and a ReLU function, respectively. It can be found from Table VI that the proposed model with each encoder block consisting of five modules attains better efficiency in terms of average recall (RE), and average F-measure (FM) against other experiments we have performed in this ablation studies. Also, the average precision (PC) value obtained by this experiment is satisfactory. From Fig. 5, it may be noticed that the model developed in the final stage of the experiment is capable of detecting moving objects accurately against other experiments.

Further, to confirm the efficacy of the suggested MSCP scheme, we compared its outcomes to recent existing methods presented in Table VII and Table VIII. These tables show that the developed technique outperformed the most recent SOTA methods in terms of accuracy on *CDnet 2014* dataset.

TABLE V
QUANTITATIVE COMPARISON WITH EXISTING METHODS (*TU-VDN*)

	Dust			Fog			Rain			Low Light		
Approaches	FM	MCC	ACC	FM	MCC	ACC	FM	MCC	ACC	FM	MCC	ACC
ViBe [85]	0.5565	0.5823	0.9740	0.6844	0.7119	0.9921	0.7307	0.7379	0.9877	0.5738	0.5954	0.9907
SuBSENSE [57]	0.5405	0.5679	0.9722	0.8204	0.8218	**0.9969**	0.6112	0.6171	0.9788	0.4812	0.5091	0.9837
LOBSTER [86]	0.4967	0.5346	0.9688	0.6649	0.6769	0.9943	0.5658	0.5949	0.9793	0.5244	0.5413	0.9890
PAWCS [55]	0.2322	0.2872	0.9656	0.5176	0.5559	0.9945	0.6628	0.6752	0.9875	0.3155	0.3505	0.9870
FST [87]	0.2360	0.2564	0.7189	0.5173	0.5659	0.9677	0.6760	0.6938	0.9892	0.4061	0.4509	0.9523
PBAS [88]	0.4033	0.4311	0.9547	0.6936	0.7086	0.9952	0.7035	0.6893	0.9823	0.5668	0.5803	0.9901
Multicue [89]	0.6166	0.6345	0.9714	0.5511	0.5913	0.9717	0.5511	0.5912	0.9717	0.4961	0.5314	0.9798
ISBM [90]	0.2191	0.2443	0.7040	0.4951	0.5456	0.9501	0.2773	0.3387	0.9312	0.3594	0.4064	0.9654
MTD [91]	0.4709	0.4772	0.9729	0.5561	0.5640	0.9928	0.4843	0.4948	0.9802	0.4656	0.4927	0.9814
VuMeter [92]	0.2171	0.2578	0.9667	0.3071	0.3698	0.6083	0.6225	0.6373	0.9814	0.3336	0.3882	0.9862
KDE [54]	0.2689	0.2978	0.9453	0.5332	0.5522	0.9938	0.6198	0.6315	0.9606	0.3845	0.3996	0.9691
MoG_V2 [93]	0.2208	0.2799	0.9673	0.2117	0.2509	0.9928	0.3532	0.3963	0.9651	0.3119	0.3486	0.9854
Eigenbackground [94]	0.3603	0.3996	0.9184	0.3413	0.3734	0.9629	0.2120	0.1761	0.6499	0.3695	0.4190	0.9673
Codebook [95]	0.2066	0.2245	0.6807	0.2319	0.3299	0.9111	0.2736	0.3958	0.9021	0.3093	0.3623	0.8848
ALWBP [65]	0.7002	0.6983	0.9763	0.7451	0.7456	0.9960	0.6962	0.6832	0.9877	0.6555	0.6658	0.9891
MSCP (Proposed)	**0.8349**	**0.8286**	**0.9864**	**0.8589**	**0.8631**	0.9929	**0.8057**	**0.8036**	**0.9934**	**0.8554**	**0.8548**	**0.9908**

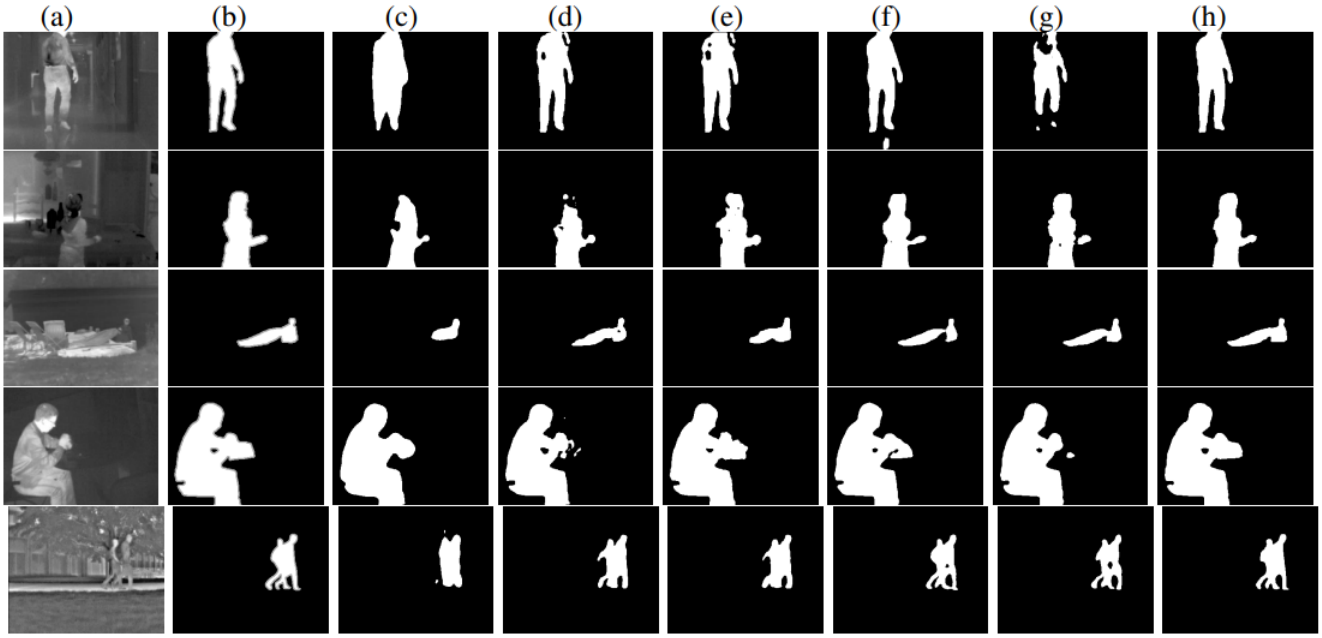

Fig. 5. Local change detection for thermal sequences of CDnet 2014 dataset: (a) Original frame (b) Ground-truth frame, Foreground masks (c) AS1, (d) AS2, (e) AS3, (f) AS4 (VGG-16), (g) AS4 (ResNet-50) and (h) **MSCP (Proposed)**, respectively.

TABLE VI
ABLATION STUDY ON FIVE THERMAL SEQUENCES (*CDnet 2014*)

Experiments	PC	RE	FM
AS1	0.8233	0.8457	0.8343
AS2	0.9199	0.8941	0.9068
AS3	0.9200	0.8918	0.9057
AS4 (VGG-16)	0.9837	0.9287	0.9554
AS4 (ResNet-50)	**0.9902**	0.8714	0.9270
MSCP (Proposed)	0.9718	**0.9858**	**0.9787**

Furthermore, to validate the efficacy of the proposed model, we assessed its performance using the *TU-VDN* dataset, specifically focusing on dynamic background scenes. From Table IX, it is found that the developed framework attained adequate accuracy for the complex dynamic background scene.

Also, the proposed scheme's statistical significance, as well as its comparison with various BS methods, is confirmed through the F-Test and detailed in Table X. In this context, $+1$, -1, and 0 indicate whether the proposed scheme performs statistically better, worse, or comparable to the BS techniques. Notably, the F-test of the developed technique yields a $+1$ and 0 value, signifying statistical superiority across all the thermal images against various SOTA techniques on the *CDnet 2014* database. Additionally, the statistical significance between the proposed scheme and various BS techniques is confirmed through the analysis of P-values on the *CDnet 2014* database and reported in Table XI. A lower P-value indicates a higher level of statistical significance for any given BS scheme. Investigation of Table XI reveals that the proposed scheme demonstrates superior statistical performance compared to the other BS techniques.

D. Unseen Video Setup

The testing and training sets contain various videos in unseen scenarios. For both training and testing image frames, there is no resemblance between the foreground and background features. We used the Leave-One-Video-Out (LOVO) strategy to separate the videos into testing and training sets, resulting in an unseen configuration. Within the context of the LOVO strategy, we assessed specific categories. For example,

TABLE VII

QUANTITATIVE ANALYSIS OF MSCP ON FIVE THERMAL SEQUENCES (*CDnet 2014* DATASET)

Techniques	Recall	Specificity	FPR	FNR	Precision	F-Measure	PWC
Autoencoder [96]	0.5938	0.9925	0.0074	0.4062	0.7311	0.6428	2.2919
RT-SBS-v2 [97]	0.8733	0.9947	0.0053	0.1267	0.8679	0.8697	1.0218
Fast BSUV-Net 2.0 [83]	0.7957	0.9951	0.0049	0.2043	0.9178	0.8379	1.2058
MSCP (Proposed)	**0.9197**	**0.9992**	**0.0009**	**0.0803**	**0.9185**	**0.9185**	**0.1177**

TABLE VIII

AVERAGE F-MEASURE COMPARISON - MSCP WITH RECENT SOTA METHODS - VARIOUS CHALLENGING SEQUENCES (*CDnet 2014*)

Category	DVTN [98]	ADNN [99]	AE-NE [100]	ZBS [101]	MU-NET [102]	GraphMOS [44]	MSCP (Proposed)
BadWeather	0.8780	0.9038	0.8337	0.9229	0.9319	0.9411	**0.9710**
Baseline	0.9811	0.9797	0.8959	0.9653	0.9875	0.9710	**0.9892**
Camera Jitter	0.9014	0.9411	0.9230	0.9545	0.9802	0.9233	**0.9932**
Dynamic Background	0.9329	0.9454	0.6225	0.9290	0.9836	0.8922	**0.9915**
Intermittent Object Motion	0.9595	0.9114	0.8231	0.8758	0.9872	0.6455	**0.9940**
Low Framerate	0.7818	**0.8123**	0.6771	0.7433	0.7237	0.6910	0.7448
Night Videos	0.7737	0.6940	0.5172	0.6800	0.8575	0.8211	**0.9660**
PTZ	0.5957	0.7424	0.8000	0.8133	0.7946	**0.8511**	0.7440
Shadow	0.9467	0.9537	0.8947	0.9765	0.9825	0.9901	**0.9955**
Thermal	0.9479	0.9411	0.7999	0.8698	**0.9825**	0.9010	0.9782
Turbulence	**0.9034**	0.8806	0.8382	0.6358	0.8499	0.8233	0.7360
Average	0.8789	0.8826	0.7841	0.8515	0.9146	0.8592	**0.9185**

TABLE IX

AVERAGE F-MEASURE AND ACCURACY ANALYSIS - MSCP - DYNAMIC BACKGROUND (DB) SCENE (*TU-VDN*)

Category	FM	ACC
DustDB	0.8165	0.9884
FogDB	0.9688	0.9995
RainDB	0.8439	0.9925
Low lightDB	0.9275	0.9952

TABLE X

QUANTITATIVE COMPARISON BETWEEN MSCP WITH SOTA METHODS USING THE F-TEST

Methods /Quantitative Measurements	PC	RE	FM	PWC
MSCP versus DeepBS [70]	+1	+1	+1	+1
MSCP versus WisenetMD [52]	+1	+1	0	+1
MSCP versus Cascade CNN [71]	+1	+1	+1	+1
MSCP versus IUTIS-5 [53]	+1	+1	0	0
MSCP versus BSUV-Net [72]	+1	0	0	+1
MSCP versus SemanticBGS [58]	+1	+1	+1	0
MSCP versus BSUV-Net 2.0 [83]	+1	0	0	0

TABLE XI

QUANTITATIVE COMPARISON BETWEEN PROPOSED SCHEME AND EXISTING TECHNIQUES USING THE P-TEST

Approaches/ Quantitative measures	PC	RE	FM	PWC
DeepBS [70]	0.0585	0.0571	0.0579	2.54e-04
WisenetMD [52]	1.83e-14	1.98e-10	2.51e-12	0.0036
Cascade CNN [71]	0.0575	5.87e-04	5.82e-04	0.0019
IUTIS-5 [53]	3.04e-15	2.12e-13	1.87e-15	8.57e-04
BSUV-Net [72]	6.29e-12	1.07e-13	5.98e-14	5.83e-04
SemanticBGS [58]	1.39e-13	1.83e-13	6.89e-15	5.71e-04
BSUV-Net 2.0 [83]	5.04e-14	1.57e-07	8.81e-09	0.0031
MSCP	**1.70e-22**	**3.27e-17**	**3.33e-19**	**1.97e-06**

in the Thermal category comprising five videos, four videos are selected for model training, while one video is chosen for model testing. Similarly, the proposed model is trained and

TABLE XII

AVERAGE F-MEASURE COMPARISON OF MSCP IN UNSEEN SETUP (*CDnet 2014*) WITH DIFFERENT TECHNIQUES (IN THIS TABLE, BL, PE, SW, BO, PA, TP, TS, BS, CO, AND T1 DEPICTS THE BLIZZARD (FROM BADWEATHER), PEDESTRIAN (FROM BASELINE), SIDEWALK (FROM CAMERA JITTER), BOATS (FROM DYNAMIC BACKGROUND), PARKING (FROM INTERMITTENT OBJECT MOTION), TURNPIKE05FPS (FROM LOW FRAMERATE), TRAMSTATION (FROM NIGHT VIDEOS), BUSSTATION (FROM SHADOW), CORRIDOR (FROM THERMAL), AND TURBULENCE1 (FROM TURBULENCE), RESPECTIVELY

Category	BL	PE	SW	BO	PA	TP	TS	BS	CO	T1
SuBSENSE [57]	0.85	0.95	0.81	0.69	0.48	0.85	0.86	0.86	0.91	**0.79**
ViBe [85]	0.53	0.90	0.30	0.22	0.26	0.60	0.67	0.67	0.75	0.58
PAWCS [55]	0.66	0.95	0.74	0.88	0.21	**0.91**	0.86	0.86	0.65	0.68
WeSamBE [56]	0.86	0.96	0.85	0.64	0.41	**0.91**	0.86	0.86	0.89	0.71
IUTIS-5 [53]	0.80	**0.97**	0.81	0.75	0.65	0.89	0.87	0.87	0.90	0.63
BSUV-Net [72]	0.82	**0.97**	0.69	0.89	0.91	**0.91**	0.80	0.94	0.83	0.66
BSUV-Net + SemanticBGS [72]	0.82	**0.97**	0.71	0.91	0.91	**0.91**	0.80	0.96	0.82	0.65
MSCP (Proposed)	**0.89**	0.95	**0.91**	**0.93**	**0.92**	0.71	**0.90**	**0.97**	**0.93**	0.69

TABLE XIII

AVERAGE F-MEASURE OF MSCP IN UNSEEN SETUP (*TU-VDN*)

	Dust (flat cluttered background)	Fog (flat cluttered background)	Rain (flat cluttered background)	Low Light (flat cluttered background)
FM	0.7605	0.7190	0.7434	0.7675
MCC	0.7513	0.7277	0.7447	0.7646
ACC	0.9294	0.9136	0.9642	0.9742

tested across various categories present in the *CDnet 2014* dataset. Table XII highlights that the proposed model achieves enhanced accuracy in comparison to SOTA techniques for the unseen configuration. Additionally, for the *TU-VDN* database, we have trained the model for dynamic background challenges under different adverse environmental conditions, including dusty, rainy, foggy, and low-light scenarios. However, the model is tested with flat cluttered background challenges under these atmospheric conditions. From Table XIII, it can be seen that the developed design attained satisfactory accuracy in all considered measures for the *TU-VDN* dataset.

V. Conclusion and Future Works

Here, we have designed an end-to-end encoder-decoder-based contrast-preserving deep neural network for thermal sequences. We propose a novel encoder network that learns deeply and extracts sparse and dense features from the input thermal sequences. The proposed multi-scale contrast preservation (MSCP) block can precisely retain the contrast details of the in-depth features and serve as a better feature representation block. The decoder network effectively classifies each pixel of the target scene as either foreground or background. The developed algorithm was tested on two benchmark databases: *CDnet 2014* and the *Tripura University Video Dataset at Night Time (TU-VDN)*. The proposed technique demonstrated better accuracy compared to thirty-eight SOTA techniques on the considered databases. It was also observed that the proposed model is more agnostic (in the case of unseen scenes) than the previous deep learning-based BSUV-Net method.

Furthermore, in this article, we developed a multi-scale contrast preservation block that can handle challenging scenes, such as deformations and changes in the appearance of moving objects, by employing a combination of multi-scale analysis, contrast preservation, and integration with the proposed encoder-decoder framework. By preserving contrast, the MSCP module facilitates the distinction between different parts of the image, even in challenging scenes where there may be deformations or changes in the appearance of the moving objects. The proposed scheme is explored on two benchmark databases: *CDnet 2014* and the *Tripura University Video Dataset at Night Time (TU-VDN)*. Within the *CDnet 2014* dataset, five thermal videos acquired by far-infrared sensors are available. These thermal videos, such as Corridor, Library, and LakeSide, present challenges, including object movements, deformations, and changes in appearance.

To assess adaptability, the developed algorithm underwent testing on these datasets, revealing its capability in handling challenges like object deformations and changes in appearance, attributed to the designed multi-scale contrast preservation (MSCP) block, which accurately retains multi-scale features with enhanced contrast. Additionally, the rainy and foggy thermal videos of the *TU-VDN* dataset also exhibit challenges such as object movements, deformations, and changes in appearance. The effectiveness of the developed model was evaluated on this dataset and was found to provide better accuracy. However, while employing multi-scale features to detect moving objects in video scenes, the proposed model lacks temporal data integration, resulting in cases where objects transitioning from movement to becoming part of the background are incorrectly classified as foreground. Furthermore, the developed model is prone to generating numerous false alarms under non-uniform lighting conditions and surface reflections.

Experiments on the two large-scale datasets (*CDnet 2014* and *TU-VDN*), showcasing a broad spectrum of challenges, demonstrate that the proposed algorithm, MSCP, effectively handles various challenging scenes, including object movements, deformations, and changes in appearance. Future improvements should optimize the scheme's effectiveness by integrating spatial and temporal details within the proposed architecture.

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